

BUS43-4

# International Economics & Globalization

Why countries trade, who wins and who loses, how exchange rates are set, and why globalization became contested

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The Economics arc meets open borders: the same micro and macro toolkit, now applied to trade, currencies, and the politics they create.

Albert School — 22 hours

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## Preface

This course closes the Economics arc you have been building since the first semester. **BUS13-4 Microeconomics** gave you the optimising agent, market equilibrium, surplus and deadweight-loss accounting, and the welfare reading of a tax. **BUS23-4 Macroeconomics** gave you the system: output, money, interest rates, the policy mix, and the way a central-bank decision propagates through an economy. **BUS33-4 Labour Economics** added the distributional turn — who gains and who loses inside a country, and the standard of causal proof modern economics

demands. International Economics takes that entire toolkit and removes the one assumption every previous course quietly kept: that the border is closed.

Opening the border changes three things at once, and the course is organised around them. First, **trade itself**: countries that can buy and sell across borders specialise, and specialisation creates gains — but those gains are never shared evenly inside a country, which is why trade is politically explosive even when it is collectively beneficial. Second, **money across borders**: once goods and capital cross frontiers, two national price levels and two interest rates must be reconciled by an **exchange rate**, and the rules governing that rate force a hard three-way choice no country can dodge. Third, **the politics**: globalization has produced a distinctive pattern of winners and losers — Milanovic’s elephant — and a new repertoire of economic statecraft, from tariffs and export controls to the weaponisation of the dollar.

We proceed in that order. Chapters 1–2 explain **why** countries trade — Ricardo’s comparative advantage, the Heckscher–Ohlin factor-proportions logic and its sharp distributional corollary (Stolper–Samuelson), then Krugman’s new trade theory, which explains the trade between **similar** countries that the classical models cannot. Chapters 3–5 turn to **trade policy**: the full welfare accounting of a tariff, the wider toolkit of quotas, subsidies and non-tariff barriers with an honest cost–benefit treatment of the arguments for protection, and then three contemporary stress tests — the US–China tariff war, the US Inflation Reduction Act, and the EU’s Carbon Border Adjustment Mechanism. Chapters 6–9 are the **open-economy macro** core: the balance of payments, exchange-rate determination (PPP in the long run, UIP in the short run), the Mundell–Fleming trilemma, and the anatomy of currency crises. Chapters 10–11 close on **globalization’s structure and politics**: global value chains and supply-chain risk, and the elephant curve with the geopolitics of reshoring, imbalances, and economic weaponisation. As throughout the arc, every abstraction is introduced by a concrete number first, and each model’s limits are stated plainly right after it.

## 1 Comparative advantage and the gains from trade

Why do countries trade at all? The deepest answer is also the most counter-intuitive: a country can gain from importing a good **even when it could produce that good more cheaply than anyone else**. The key is not who is better in an absolute sense, but who gives up **less** to produce one good rather than another — the **opportunity cost** you met on day one of Microeconomics, now applied across borders.

### 1.1 Absolute versus comparative advantage

**Worked example — the lawyer and the typist.** A lawyer types twice as fast as her assistant **and** argues cases far better. She has an absolute advantage at both tasks. Should she type her own briefs? No — every hour she spends typing is an hour not spent in court, where her edge is enormous. The hour she “saves” by typing costs her a far more valuable hour of legal work. She should specialise in law and let the assistant type, even though the assistant is the worse typist. Countries face exactly this logic: what matters is the **opportunity cost**, not absolute productivity.

Make it numerical. Two countries, **Home** and **Foreign**, each with labour as the only input, producing **cloth** and **wine**. The table gives unit labour requirements — hours needed to make one unit:

|         | <b>Cloth</b> ( $a_{LC}$ ) | <b>Wine</b> ( $a_{LW}$ ) |
|---------|---------------------------|--------------------------|
| Home    | 1 hr                      | 2 hr                     |
| Foreign | 6 hr                      | 3 hr                     |

Home needs fewer hours for **both** goods — it has an absolute advantage in each. Yet trade still pays, because the **opportunity costs** differ. At Home, making one cloth means giving up the wine those labour-hours could have made:  $a_{LC}/a_{LW} = 1/2$  a unit of wine. In Foreign the same cloth costs  $6/3 = 2$  units of wine. Cloth is **cheap** at Home in the only sense that matters for trade — Home sacrifices less wine to make it.

**Definition 1** A country has a **comparative advantage** in the good it can produce at lower **opportunity cost** — i.e. by giving up less of the other good — regardless of absolute productivity. With one good and two countries, comparative advantage is always symmetric: if Home's opportunity cost of cloth is lower, then Foreign's opportunity cost of **wine** is necessarily lower, so each country has a comparative advantage in exactly one good. The autarky relative price of cloth equals its opportunity cost:  $(P_C/P_W)^{\text{Home}} = 1/2$ ,  $(P_C/P_W)^{\text{Foreign}} = 2$ .

## 1.2 Gains from specialisation and trade

Because cloth is relatively cheap at Home ( $1/2$ ) and dear in Foreign ( $2$ ), a **world relative price** of cloth that settles anywhere strictly between  $1/2$  and  $2$  lets **both** countries gain. Suppose it settles at  $P_C/P_W = 1$ : one cloth trades for one wine on the world market.

**Worked example — both countries beat autarky.** Take a world price of 1 wine per cloth. Home specialises entirely in cloth. With (say) 100 hours it makes 100 cloth, keeps 60, and exports 40 cloth for 40 wine. In autarky those last 40 cloth would have bought only  $40 \times 1/2 = 20$  wine — trade **doubles** the wine Home gets for its cloth. Foreign specialises in wine: each cloth it imports costs 1 wine instead of the 2 wine it would have sacrificed making cloth at home. Nobody produced more **hours**; the world simply re-allocated each country's labour to what it does comparatively best, and total output of both goods rose.

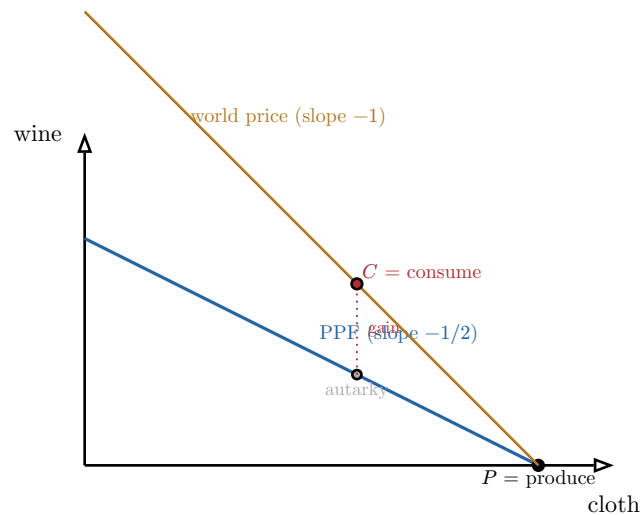


Figure 1: Gains from trade for Home. Producing only cloth (point  $P$ ) and trading at the world price (the steeper line is the budget the world market offers, steeper than the PPF because cloth is worth more abroad) lets Home consume at  $C$ , **outside** its own production frontier. The vertical gap over the best autarky point is the gain from trade — extra consumption from no extra labour, purely by specialising where Home’s opportunity cost is lowest.

**Common pitfall** The single most common error in this whole course is to argue from **absolute** productivity — “Foreign is less productive at everything, so it can’t compete and trade must hurt it.” Wrong on both counts. Foreign still has a comparative advantage in **something** (here, wine), and it gains by specialising there. A country is never “uncompetitive at everything”: comparative advantage is defined by **internal** opportunity-cost ratios, which always favour one good. Low absolute productivity shows up in low **wages**, not in an inability to trade.

### 1.3 Heckscher–Ohlin: trade from factor proportions

Ricardo explains **that** opportunity costs differ but treats the difference as given. Heckscher and Ohlin explain **where it comes from**: countries differ in their **endowments** of factors (capital, skilled and unskilled labour, land), and goods differ in the **intensity** with which they use those factors.

**Definition 2** The **Heckscher–Ohlin theorem**: a country exports the good that uses **intensively** the factor it is **abundantly** endowed with, and imports the good using its scarce factor intensively. A capital-abundant country exports capital-intensive goods; a labour-abundant country exports labour-intensive goods. Trade is, in effect, an indirect way of exporting a country’s **abundant factor** (embodied in goods) and importing its **scarce** one.

### 1.4 Stolper–Samuelson: who wins and who loses inside a country

Heckscher–Ohlin’s real payoff is distributional, and it is the hinge of the whole course. If trade raises the relative price of the good using the abundant factor, it must raise the **real return to that factor** — and lower the real return to the scarce factor.

**Worked example — the rich-country production worker.** An advanced economy is abundant in **skilled** labour and scarce in **unskilled** labour. Opening to trade with labour-

abundant economies raises the relative price of skill-intensive goods (which it exports) and lowers that of labour-intensive goods (which it now imports). Skilled wages rise; **unskilled** wages fall — not just relative to skilled, but in real terms. The stagnating wages of less-educated workers in rich countries since the 1980s are exactly what this theorem predicts, and exactly why trade is politically contested even when the nation as a whole gains.

**Theorem 3 (Stolper–Samuelson)** A rise in the relative price of a good raises the real return of the factor used **intensively** in producing it and lowers the real return of the other factor — **unambiguously**, not just in relative terms. Trade liberalisation, which moves domestic prices toward world prices, therefore creates **real winners and real losers within each country**: the abundant factor gains, the scarce factor loses, even though the country's total income rises.

**Common pitfall** The nation gaining and **every** citizen gaining are different statements. The gains-from-trade result (Section 1.2) says the winners' gains **exceed** the losers' losses, so winners **could** compensate losers and all be better off. Whether they **do** compensate — through redistribution, retraining, transfers — is a **policy** choice, not a property of trade. Skip the compensation and a trade deal that raises GDP can still leave a large, identifiable, and angry group genuinely worse off. This gap between potential and actual compensation is where the politics of Chapters 5 and 11 lives. (This is the distributional question of **BUS33-4**, now driven by trade rather than technology.)

## 2 New trade theory: scale, variety, and geography

The classical models share an awkward silence. They predict that **different** countries trade **different** goods — capital-rich exporting machines for the labour-rich country's textiles. Yet the largest trade flows in the world are between **similar** rich countries exchanging **similar** goods: Germany sells cars to France and France sells cars to Germany. Comparative advantage cannot explain this. Krugman's new trade theory can.

### 2.1 Increasing returns and intra-industry trade

**Worked example — why France and Germany swap cars.** France and Germany have near-identical factor endowments, so Heckscher–Ohlin predicts little trade between them — yet they are each other's biggest trading partners, and much of it is **cars for cars**. The reason is not comparative advantage but **scale and variety**. A car model is cheaper per unit the more of it you make (increasing returns), so no country can profitably make every variety. Each specialises in a few models, and consumers everywhere get the whole range. Renault and Volkswagen both gain larger markets; French and German drivers both gain more choice.

**Definition 4** **Intra-industry trade** is the two-way exchange of **differentiated varieties** within the same industry, driven by **economies of scale** and consumers' preference for **variety**, not by factor endowments. In Krugman's monopolistic-competition model, opening to trade enlarges the market: each firm produces a longer run of fewer varieties (moving down its average-cost curve), so prices **fall** and the **number of varieties available** rises. The gains are **lower prices and more choice**, and they accrue even between identical countries.

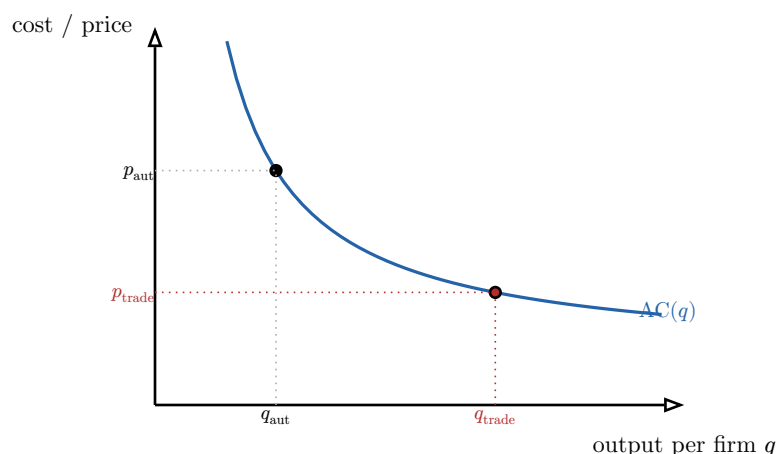


Figure 2: Economies of scale and the gains from intra-industry trade. Each surviving firm's average cost falls as its output rises. Opening trade enlarges the market, so each firm produces a longer run  $q_{\text{trade}} > q_{\text{aut}}$  and slides down its AC curve — prices fall from  $p_{\text{aut}}$  to  $p_{\text{trade}}$ . Fewer firms each serve a bigger market, but because the **combined** market supports more firms than either national market alone, consumers also get **more varieties**. Gains appear even with identical countries and no comparative advantage.

## 2.2 Economic geography and the home-market effect

**Remark** Scale economies also explain **where** industries locate. If production is cheaper the more is made in one place, activity **agglomerates**: a region with a head start attracts more firms, which attract more workers, which enlarge the local market and attract still more firms — a self-reinforcing cluster (Silicon Valley, the City of London, the German auto belt). The **home-market effect** is the corollary: countries tend to become **net exporters** of goods for which they have a large domestic market, because scale makes large-market location efficient. Geography, not just endowments, shapes the pattern of trade.

**Common pitfall** New trade theory complicates the welfare verdict in two ways. First, because trade here is driven by scale rather than factor reallocation, its **distributional** bite is softer than Stolper–Samuelson's — adjustment is more about which **firms** survive than which **factor** loses, so intra-industry trade is politically easier than inter-industry trade. Second, with increasing returns and imperfect competition, the textbook case for **pure** free trade weakens: there is a theoretical opening for **strategic trade policy** (subsidising a national champion to capture a scale-driven global market). That opening is real but narrow — it needs information governments rarely have and invites retaliation, as Chapter 4 makes precise.

## 3 The welfare economics of a tariff

A tariff is just a tax on imports, and you already know how to read a tax: it is the surplus accounting of **BUS13-4**, applied at the border. The novelty is that a tariff redistributes among **four** parties — domestic consumers, domestic producers, the government, and (for a large country) foreigners — and the national verdict is the sum of those pieces.

### 3.1 The small-country case: a pure efficiency loss

A **small** country takes the world price as given (its imports are too small to move it). Consider a market with domestic demand and supply, a world price  $P_w = 2$ , and a tariff  $t = 2$  that raises the internal price to  $P_t = 4$ .

**Worked example — a tariff, fully accounted.** Demand is  $P = 10 - Q$  and supply is  $P = Q$  (quantities in thousands of units, prices in €). At the world price  $P_w = 2$ : domestic firms supply  $Q_S = 2$ , consumers buy  $Q_D = 8$ , so imports are 6. Impose the €2 tariff; the internal price rises to 4. Now  $Q_S = 4$ ,  $Q_D = 6$ , imports shrink to 2. The four effects:

- **Producer surplus** rises by area  $a = \frac{1}{2}(2 + 4)(4 - 2) = \text{€}6\text{k}$ .
- **Government revenue** is  $c = t \times \text{imports} = 2 \times 2 = \text{€}4\text{k}$ .
- **Consumer surplus** falls by  $a + b + c + d = \frac{1}{2}(8 + 6)(4 - 2) = \text{€}14\text{k}$ .
- The two triangles  $b = \frac{1}{2}(4 - 2)(2) = \text{€}2\text{k}$  and  $d = \frac{1}{2}(8 - 6)(2) = \text{€}2\text{k}$  are **pure deadweight loss** — surplus that goes to nobody.

National welfare change  $= a + c - (a + b + c + d) = -(b + d) = -\text{€}4\text{k}$ . The producers' and government's gains are **transfers** from consumers; the €4k of triangles is the genuine national loss.

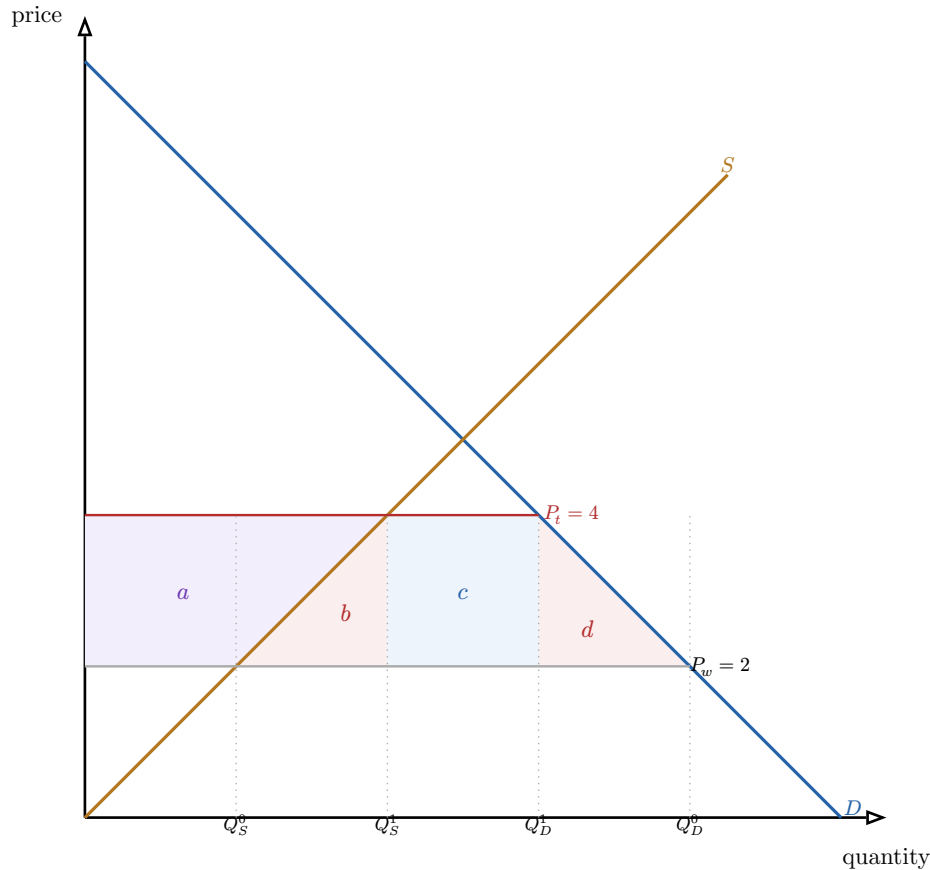


Figure 3: The welfare anatomy of a tariff (small country). The internal price rises from  $P_w$  to  $P_t$ . Consumers lose  $a + b + c + d$ . Of that,  $a$  is transferred to producers and  $c$  to the government as tariff revenue — pure redistribution. The two red triangles  $b$  (production distortion: resources pulled into high-cost domestic output) and  $d$  (consumption distortion: valuable trades that no longer happen) go to **no one**: they are the **deadweight loss**, the country's net welfare cost.

**Definition 5** For a **small** (price-taking) importer, a tariff produces:

- a **consumer-surplus loss** equal to  $a + b + c + d$ ;
- a **producer-surplus gain**  $a$  and **tariff revenue**  $c$  — both transfers **from** consumers, not new wealth;
- a **net national welfare loss** equal to the two deadweight triangles,

$$DWL = b + d,$$

where  $b$  is the **production-distortion** loss (output made at home above world cost) and  $d$  is the **consumption-distortion** loss (consumers priced out of valuable purchases). For a small country a tariff is **unambiguously** welfare-reducing.

### 3.2 The large-country case: the terms-of-trade gain

A country large enough that cutting its imports **depresses the world price** can do something a small country cannot: shift part of the tariff's burden onto foreigners.

**Definition 6** A **large** importing country, by restricting demand, lowers the world price its suppliers receive — a **terms-of-trade gain** (call its value  $e$ ). Its net welfare change is

$$\Delta W = \underbrace{e}_{\text{terms-of-trade gain}} - \underbrace{(b + d)}_{\text{efficiency loss}}.$$

If  $e > b + d$  the tariff **raises** national welfare. The tariff that maximises  $\Delta W$  is the **optimal tariff**; it is strictly positive for a large country.

**Common pitfall** The optimal-tariff argument is theoretically valid and practically dangerous. First, the terms-of-trade gain is **extracted from trading partners** — it is a beggar-thy-neighbour policy that lowers **world** welfare even when it raises the imposing country's. Second, it **invites retaliation**: if partners impose their own optimal tariffs, everyone ends up with efficiency losses and **no** terms-of-trade gain — the prisoner's-dilemma structure (recall the payoff matrices of **BUS13-4**) that the whole multilateral trading system exists to escape. "A tariff can help a large country" is true in isolation and false once the other player moves.

## 4 Beyond tariffs: barriers, subsidies, and the case for protection

Tariffs are the most legible instrument, but most modern protection works through other channels. Each can be mapped back to the same surplus accounting — and the classic arguments for protection each survive a rigorous cost-benefit test only under conditions worth stating precisely.

### 4.1 Quotas, subsidies, and non-tariff barriers

**Definition 7**

- A **quota** caps the physical quantity imported. It raises the internal price just as a tariff does, with the same consumer loss and deadweight triangles — but the rectangle  $c$  becomes **quota rents** captured by whoever holds the import licences (foreign exporters,

if quotas are allocated abroad) rather than government revenue. A quota and a tariff are “equivalent” only when the government auctions the licences.

- An **export subsidy** pays domestic firms to sell abroad. It raises the internal price (goods are diverted to export), costs the treasury the subsidy, and generates deadweight loss **plus** a terms-of-trade **loss** — the country sells to foreigners below cost. It is the mirror image of the optimal tariff: an unambiguous national loss.
- **Non-tariff barriers** (NTBs) — technical standards, sanitary rules, licensing, customs friction, local-content requirements — restrict imports without a visible tax. They are the dominant form of modern protection precisely because they are **opaque** and hard to challenge legally, even though their welfare effects mirror those of tariffs and quotas.

## 4.2 The arguments for protection, costed

**Definition 8** Three protectionist arguments recur; each is conditionally valid.

- **Infant industry**: a young industry with steep learning curves or scale economies may be unprofitable now but viable later; temporary protection lets it grow. **Valid only if** the future gains exceed the present deadweight cost **and** the firm could not simply borrow against those future profits — and it requires a credible **exit**, which politics rarely delivers.
- **Reciprocity**: open only as much as partners do. A bargaining lever in trade negotiations, but as an economic principle it confuses the **means** (mutual opening) with the **end** (a country gains from its **own** liberalisation regardless of others).
- **National security**: some capacity (food, defence, now semiconductors and critical minerals) is worth protecting at an efficiency cost because reliance on a hostile supplier is a strategic risk. **Valid** where the security externality is real — but it is the argument most easily **abused**, since any industry can dress itself as strategic.

## 4.3 The WTO and its limits

**Remark** The **WTO** institutionalises the escape from the tariff prisoner’s dilemma: members bind tariffs at negotiated ceilings, extend any concession to all members (**most-favoured-nation** non-discrimination), and submit disputes to binding adjudication. It works well for **tariffs on goods**. Its limits are now acute: it struggles with **NTBs and subsidies** (hard to define and verify), with **services and digital** trade, and with **state-capitalist** industrial policy; and since 2019 its **Appellate Body** has been paralysed by the US blocking new judges, leaving the dispute system without a final court. The rules-based order of Chapter 3’s logic is, in practice, fraying — which is the backdrop to the stress tests next.

## 5 Trade policy under stress: three contemporary episodes

The tools of Chapters 3–4 are not museum pieces; they are exactly what is needed to read the trade conflicts of the last decade without retreating to ideology. Each episode below is an application of **named winners and losers on course-derived mechanisms** — the analytical skill this course is built to deliver.

## 5.1 The US–China tariff war, 2018–2025

**Worked example — who paid for the China tariffs?** From 2018 the US imposed tariffs rising to 25% on hundreds of billions of dollars of Chinese imports; China retaliated on US farm and industrial goods. Run the small-country accounting. Empirical studies (the tariff **pass-through** was found to be near-complete) show internal US prices rose by close to the full tariff — meaning the burden fell on **US consumers and import-using firms** (area  $a + b + c + d$ ), not on Chinese exporters. US import-competing producers gained  $a$ ; the Treasury collected  $c$ ; the triangles  $b + d$  were lost. China’s retaliation hit US **farmers**, who then required federal bailouts — a transfer **back** to the losers, financed by taxpayers. Net: a textbook small-country welfare loss on both sides, with a sharp internal redistribution and trade **diverted** (not eliminated) toward Vietnam and Mexico.

## 5.2 The US Inflation Reduction Act (2022)

**Worked example — subsidy as industrial policy.** The IRA offers large **production and consumer subsidies** for clean-energy manufacturing sited in North America (local-content rules on EV batteries, tax credits for domestic production). Read it with the course’s tools: it is a **production subsidy plus a local-content NTB**. Domestic winners are US clean-energy producers and workers (and the **climate externality**, a genuine welfare gain absent from the pure-trade accounting). Losers are the US **treasury**, and **foreign** producers — notably the EU and Korea — discriminated against by the local-content rules, who decry it as a subsidy race. It is the **strategic-trade** and **national-security** arguments of Chapter 4 made policy, with a real externality justification **and** a real beggar-thy-neighbour edge.

## 5.3 The EU Carbon Border Adjustment Mechanism

**Worked example — a tariff that internalises an externality.** The CBAM charges importers of carbon-intensive goods (steel, cement, aluminium, fertiliser) a levy equal to the EU carbon price their foreign producers did not pay. Formally it is a **tariff**; its **justification** is the Pigouvian logic of **BUS13-4**. Without it, the EU’s own carbon price would simply cause **carbon leakage** — production and emissions relocating to unregulated countries, and EU firms undercut by dirtier imports. CBAM levels the field and prices the externality at the border. Winners: EU producers (no longer undercut) and the **climate**. Losers: carbon-intensive exporters to the EU (notably some emerging economies), who call it green protectionism — and whose objection that it burdens **development** is the distributional politics of Chapter 11 in advance.

**Common pitfall** Notice how the **same instrument** earns opposite verdicts depending on the externality. A tariff with no externality (US–China) is a net loss disguising redistribution. A tariff **correcting** an externality (CBAM) can raise welfare — exactly the Pigouvian exception. The analyst’s job is never to be “for” or “against” tariffs but to **name the mechanism, the externality (if any), and the winners and losers**, then add up the surplus. Ideology substitutes a slogan for that accounting.

## 6 The balance of payments

We now cross from goods to money. Every cross-border transaction — a car exported, a bond bought, a dividend paid — is recorded in the **balance of payments**. Reading it is the open-

economy analogue of the national accounts you met at the end of **BUS13-4** and used throughout **BUS23-4**.

## 6.1 Construction and the accounting identity

**Definition 9** The **balance of payments** records all transactions between residents and the rest of the world over a period, in two main accounts:

- the **current account** (CA): trade in goods and services, primary income (cross-border wages and investment income), and secondary income (transfers);
- the **financial account** (FA): cross-border purchases of assets — direct investment, portfolio flows, reserves.

Built on **double-entry** bookkeeping, every transaction is recorded twice (a credit and a debit), so the accounts must sum to zero. Ignoring the small capital account and statistical discrepancy,

$$CA + FA = 0 \quad \text{i.e.} \quad CA = -FA.$$

**Worked example — a deficit is a capital inflow.** Suppose a country imports €100bn more than it exports — a €100bn current-account **deficit**. Those imports were paid for somehow: foreigners ended up holding €100bn of the country's assets (bonds, equities, deposits) — a €100bn financial-account **surplus** (capital inflow). The two are not a coincidence; they are the **same event** seen from two sides. A current-account deficit is, by identity, a net **sale of assets to** (or **borrowing from**) the rest of the world.

## 6.2 Interpreting imbalances

**Common pitfall** A current-account deficit is **not** automatically a sign of weakness, nor a surplus a sign of strength. The identity  $CA = S - I$  (a country's external balance equals national saving minus investment) shows a deficit can mean either **healthy** investment exceeding domestic saving (productive borrowing, as in a fast-growing economy) or a **savings shortfall** (a country living beyond its means). Likewise a surplus can reflect **competitive** exporting or **deficient domestic demand** (Germany's surplus, discussed in Chapter 11, is partly the latter). The number is a **question**, not a verdict — exactly as GDP was in **BUS13-4**. What matters is **why** saving and investment diverge, and whether the resulting external position is sustainable.

## 7 Exchange rates: PPP in the long run, UIP in the short run

What determines the price of one currency in terms of another? Two forces, on two horizons. In the **long run** the exchange rate is anchored by goods prices — **purchasing power parity**. In the **short run** it is driven by capital chasing returns — **uncovered interest parity**. Holding the two horizons apart is the key to the whole chapter.

### 7.1 Purchasing power parity (the long run)

**Worked example — the Big Mac test.** A Big Mac costs \$5.00 in the US and £4.00 in the UK. If the burger is the same good, it should cost the same once converted — so the implied exchange rate is  $5.00/4.00 = 1.25$  dollars per pound. If the **market** rate is instead \$1.40 per pound, the pound is “overvalued” by PPP: £4 of burger costs \$5.60 at market rates but only

\$5.00 of US burger. PPP says such gaps should erode over time, as cheap-country goods get bought up and dear-country goods shunned.

**Definition 10 Absolute PPP** says the nominal exchange rate equates national price levels: with  $E$  the price of foreign currency in home currency,

$$E = P/P^*,$$

where  $P, P^*$  are the home and foreign price levels. **Relative PPP** — the empirically usable version — says the **rate of change** of  $E$  tracks the **inflation differential**:

$$\frac{\Delta E}{E} \approx \pi - \pi^*.$$

A country with persistently higher inflation sees its currency **depreciate** at roughly the inflation gap. PPP is the **long-run** anchor; it fails badly year-to-year (many goods are non-traded, and arbitrage is slow) but holds far better over decades.

## 7.2 Uncovered interest parity (the short run)

**Worked example — where does the hot money go?.** A euro deposit pays 3%; a dollar deposit pays 5%. Does money flood to dollars? Not necessarily. If investors expect the dollar to **depreciate** by 2% against the euro over the year, the extra 2% interest is exactly wiped out by the 2% currency loss — leaving the two deposits equally attractive. In equilibrium the interest advantage of a currency is offset by its **expected depreciation**. That balancing condition pins down today's exchange rate given expectations.

**Definition 11 Uncovered interest parity (UIP):** the home interest rate equals the foreign rate plus the expected depreciation of the home currency,

$$i = i^* + \frac{E^e - E}{E},$$

where  $E^e$  is the expected future exchange rate. Capital flows enforce it: any uncompensated interest gap would be arbitrated away. UIP makes the exchange rate an **asset price**, set in the short run by interest differentials and expectations — which is why a surprise central-bank rate hike (the monetary transmission of **BUS23-4**) moves the currency **immediately**, long before any goods price adjusts.

**Common pitfall** Keep the two horizons separate. PPP and UIP can point in **opposite** directions at once: a country that raises interest rates sharply will, by UIP, see its currency **appreciate** now even if high inflation means PPP predicts long-run **depreciation**. This is **overshooting** (Dornbusch): because goods prices are sticky but the asset market clears instantly, the exchange rate jumps **past** its long-run value and then drifts back. Confusing the short-run asset-market story with the long-run goods-market anchor is the classic error in reading currency movements.

## 8 The Mundell–Fleming trilemma

Open-economy macro delivers one inescapable constraint, and it is the single most useful idea in this half of the course. A country simply **cannot have all three** of: a fixed exchange rate, free movement of capital, and an independent monetary policy. It must give one up.

### 8.1 Why you can only pick two

**Worked example — the impossible peg.** Suppose a country fixes its currency to the dollar, lets capital move freely, **and** tries to cut interest rates to fight a recession. The moment its rate falls below the US rate, UIP bites: investors sell the currency for higher-yielding dollars, capital floods out, and the peg breaks. To **hold** the peg with open capital markets, the country must keep its rate equal to the anchor's — i.e. surrender monetary independence. The three goals are mutually inconsistent; any two can coexist only by sacrificing the third.

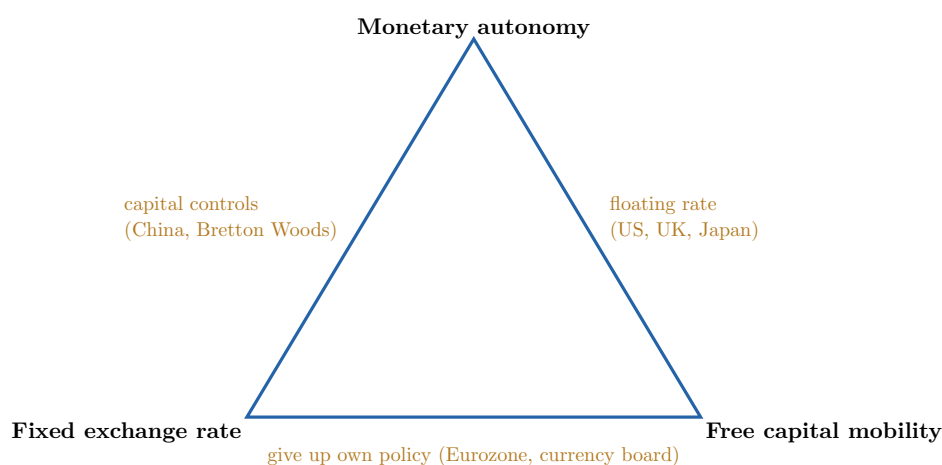


Figure 4: The Mundell–Fleming trilemma. Each **vertex** is a policy goal; each **edge** is a feasible regime that attains the two goals it connects by sacrificing the opposite vertex. Capital controls keep a fixed rate **and** monetary autonomy by closing the capital account; a floating rate keeps autonomy **and** open capital by letting the currency move; a currency union or currency board keeps a fixed rate **and** open capital by surrendering monetary policy. The interior is unreachable — you cannot have all three.

### 8.2 The three corners in the real world

**Definition 12** The trilemma classifies real regimes by the corner each sacrifices:

- the **Eurozone** gives up **national monetary autonomy** (one ECB rate for all members) to keep fixed intra-area rates and open capital — which is why a country in recession cannot ease on its own, the design flaw exposed in 2010–12;
- **China** long gave up **free capital mobility** (capital controls) to manage the **RMB** while running its own monetary policy, and liberalises the capital account only cautiously for exactly this reason;
- most **emerging economies** and advanced floaters give up the **fixed rate**, accepting currency volatility to keep monetary independence with open markets.

Every exchange-rate-regime debate is, at bottom, an argument about **which corner to sacrifice**.

## 9 Currency crises

When a country tries to hold an exchange-rate commitment it cannot back, the result is a **currency crisis** — a sudden, often violent collapse of a peg. The literature distinguishes three “generations” of model, each written to explain a crisis the previous generation could not.

### 9.1 First generation: fundamentals (Krugman)

**Worked example — draining the reserves.** A government pegs its currency but runs a chronic budget deficit financed by printing money. The rising money supply is inconsistent with the peg, so the central bank must sell foreign-currency reserves to defend it. Speculators can see the reserves falling toward zero on a predictable date — so they attack **just before** exhaustion, dumping the currency all at once and forcing the collapse early. The crisis is the inevitable consequence of **bad fundamentals**.

**Definition 13 First-generation** (Krugman, 1979) models: an exchange-rate peg is undermined by **inconsistent fundamentals** — typically monetised fiscal deficits draining reserves. The collapse is **predictable** and follows from policy contradictions; the speculative attack merely brings forward the unavoidable. Cause: the government’s own incompatible policies.

### 9.2 Second generation: self-fulfilling expectations (Obstfeld)

**Worked example — the peg that breaks because people expect it to.** A country has **defensible** fundamentals but defends its peg with high interest rates that worsen unemployment. If markets believe the government will tolerate the pain, the peg holds. If markets **doubt** that resolve, they attack; defending now requires **even higher** rates, raising the political cost of staying pegged until the government rationally abandons it — **validating** the attack. The same fundamentals support **two** equilibria; expectations select which. Britain’s 1992 ERM exit fits this mould.

**Definition 14 Second-generation** (Obstfeld) models: a crisis is **self-fulfilling**. The government weighs the cost of defending the peg against the cost of abandoning it; because defence becomes costlier when markets expect collapse, there are **multiple equilibria**, and a shift in expectations alone can trigger a crisis with **no deterioration in fundamentals**. Cause: a coordination failure among speculators.

### 9.3 Third generation: balance sheets and twin crises

**Worked example — Asia, 1997.** East Asian banks and firms borrowed heavily in **dollars** but earned in **local currency** — a currency mismatch. When the Thai baht’s peg broke and the currency fell, the local-currency value of that dollar debt **exploded**, bankrupting firms and banks at once. Currency collapse and banking collapse fed each other — a **twin crisis** — and contagion spread the panic across the region. Fundamentals looked sound by first-generation standards (budgets were near balance); the rot was in private **balance sheets**.

**Definition 15 Third-generation** models: crises driven by **financial-sector fragility** — **currency mismatch** (borrowing in foreign currency, earning in domestic), maturity

mismatch, and weak balance sheets — in which currency depreciation and banking collapse are mutually reinforcing (**twin crises**), amplified by **contagion**. The 1997 Asian crisis is the canonical case. Cause: balance-sheet vulnerability in the private financial system.

**Worked example — Argentina, 2001.** Argentina pinned the peso one-to-one to the dollar via a **currency board** (Chapter 8’s “give up monetary autonomy” corner), which tamed hyperinflation but left the peso badly overvalued. Public and private debt was in dollars; with no ability to devalue or print, adjustment fell entirely on wages and output through brutal deflation and recession. As reserves and political tolerance ran out, the board collapsed in 2001–02: the peg broke, the dollar debt’s peso value exploded, and the country defaulted. Argentina blends all three generations — fiscal strain (first), a credibility collapse (second), and dollarised balance-sheet mismatch (third).

**Common pitfall** The three generations are **complements, not rivals**. Real crises usually mix them: weak fundamentals (first) make a country **vulnerable** to a self-fulfilling attack (second), which detonates **balance-sheet** mismatches (third). The diagnostic question is not “which model is correct?” but “which mechanism **dominates** this episode?” — and the trilemma tells you **why the country was exposed** in the first place: it had taken on a fixed-rate commitment with open capital markets that its fundamentals or balance sheets could not back.

## 10 Global value chains and supply-chain risk

Modern goods are not made in one country; they are assembled from tasks scattered across many. A smartphone is designed in California, runs on chips fabricated in Taiwan with Dutch lithography, and is assembled in China from components made across Asia. Measuring and reading these **global value chains** (GVCs) requires data on **value added**, not gross trade — and exposes new kinds of risk.

### 10.1 Trade in value added (TiVA)

**Worked example — who really makes the phone?.** A \$1,000 phone assembled in China and shipped to the US shows up as a \$1,000 Chinese **export** in gross trade statistics. But China may add only \$50 of assembly value; the chips, screen, software, and design value accrue to Taiwan, Korea, the Netherlands, the US. Gross trade **triple-counts** components that cross borders several times and badly misattributes **who captures the value**. The OECD’s **Trade in Value Added** (TiVA) database re-allocates each export to the countries that actually generated the value.

**Definition 16** **Trade in value added** decomposes a country’s gross exports into **domestic value added** (value generated at home) and **foreign value added** (imported inputs embodied in the export). Two structural measures follow:

- **backward linkage**: the foreign-value-added share of exports — how much a country **depends on imported inputs** to produce its own exports;
- **forward linkage**: how much of a country’s value added is embodied in **other countries’** exports — how much others depend on it.

**Dependency mapping** uses these to locate **chokepoints**: a single supplier nation or firm on which many downstream producers rely.

## 10.2 Supply-chain stress under geopolitical shock

**Remark** GVCs trade **efficiency** (each task done where it is cheapest, lean “just-in-time” inventories) for **fragility** (a break anywhere halts the chain). Three recent shocks made the trade-off vivid:

- **COVID-19** (2020–22): factory shutdowns and a demand whiplash caused cascading shortages — semiconductors above all — revealing how thin just-in-time buffers were;
- the **Ukraine war** (2022–): exposed European dependence on Russian **energy** and on Ukrainian/Russian grain, neon, and metals — a concentrated-supplier risk realised;
- **semiconductor export controls** (US restrictions on advanced chips and the tools to make them to China, from 2022): turned a **chokepoint** — the handful of firms and countries that make leading-edge chips and lithography — into an instrument of **policy**.

The policy response — **reshoring** and **friendshoring** (Chapter 11) — buys resilience by sacrificing some of the comparative-advantage gains of Chapter 1. That is a real cost, not a free lunch.

## 11 The distributional politics of globalization

We close where the course began — with **who wins and who loses** — but now at the **global** scale, and with the geopolitics that the pattern has produced. The Stolper–Samuelson losers of Chapter 1 turn out to be visible in a single, now-famous picture.

### 11.1 Milanovic’s elephant curve

**Worked example — two stagnant decades, two very different reasons.** Between 1988 and 2008 the **Chinese factory worker** saw real income roughly double, lifted by exactly the trade integration this course describes. Over the **same** twenty years the **American or French lower-middle-class worker** saw almost no real gain. And the **global top 1%** pulled far ahead. Plot real income growth against position in the **global** income distribution and these three facts trace the silhouette of an elephant — the single most important picture in the politics of globalization.

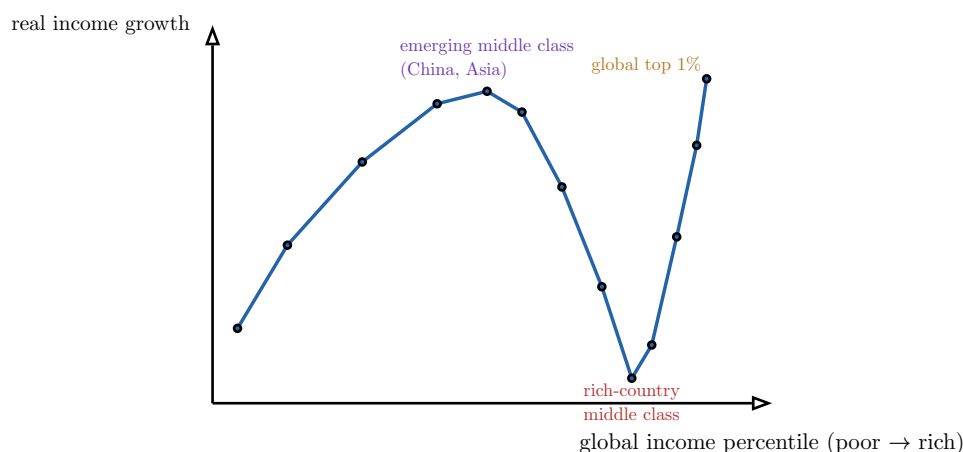


Figure 5: Milanovic’s **elephant curve** (income growth 1988–2008 by global percentile, schematic). The **body’s hump** around the global median — the emerging middle class of Asia — enjoyed the fastest growth. The **trough** near the 80th percentile — the **lower-middle class of rich countries** — barely gained: globalization’s relative losers, and the Stolper–Samuelson scarce factor of Chapter 1. The **raised trunk** at the very top — the global elite — captured a disproportionate share. The shape **is** the politics of trade.

**Definition 17** The **elephant curve** (Lakner–Milanovic) plots real income growth against position in the **global** income distribution over the era of hyper-globalization. Its three features map to three political constituencies: the **emerging middle class** (large gains, the hump), the **rich-country lower-middle class** (stagnation, the trough), and the **global top 1%** (large gains, the trunk). The trough is the empirical face of the Stolper–Samuelson loser, and explains why the **backlash** against trade is concentrated in rich-country electorates even as global inequality **between** countries fell.

## 11.2 Imbalances, reshoring, and economic weaponisation

**Definition 18** The same forces produced a distinctive **geopolitical** configuration:

- **global imbalances**: persistent **surplus** economies (Germany, China — high saving, suppressed domestic demand) mirrored by persistent **deficit** economies (notably the US), the  $CA = S - I$  identity of Chapter 6 written across decades, and a recurring source of friction;
- **reshoring and friendshoring**: relocating production home or to allied countries to cut the GVC dependency of Chapter 10 — buying resilience at the cost of comparative-advantage efficiency, and fragmenting the world into trading blocs;
- **economic weaponisation**: the use of economic interdependence as **coercion** — **financial sanctions** (cutting Russia from dollar clearing and SWIFT, freezing central-bank reserves), **dollar primacy** (the leverage the US gains from the dollar’s role in global payments), and **semiconductor export controls** (denying an adversary the chokepoint technology of Chapter 10).

**Common pitfall** Weaponising interdependence is powerful but **self-eroding**. Each use of the dollar system or a technology chokepoint as a weapon gives targets a reason to **build alternatives** — non-dollar payment rails, indigenous chip industries, diversified reserves.

The very centrality that makes the tool effective today is what its use teaches rivals to dismantle tomorrow. As with the optimal tariff of Chapter 3, the static gain ignores the **strategic response** — and the long-run cost of fragmentation is paid in the lost gains-from-trade of Chapter 1.

**The thread of this book.** Open the border and the Economics toolkit explains three things at once. **Trade:** countries specialise by **comparative advantage** (Ricardo) — sourced from **factor endowments** (Heckscher–Ohlin) with a sharp distributional sting (**Stolper–Samuelson:** the scarce factor loses in real terms) — while **new trade theory** (Krugman) adds the scale-and-variety trade between **similar** countries. **Trade policy** is surplus accounting at the border: a tariff is a deadweight loss  $b + d$  wrapping transfers, redeemed only by a **terms-of-trade** gain (large country) or a corrected **externality** (CBAM), and the US–China war, the IRA, and CBAM are this accounting applied to live conflicts. **Open-economy money:** the **balance of payments** identity ( $CA = -FA = S - I$ ); exchange rates anchored by **PPP** in the long run and **UIP** in the short; the **trilemma** forcing every country to sacrifice one of fixed rate, open capital, monetary autonomy; and **currency crises** in three reinforcing generations (fundamentals, self-fulfilling, balance-sheet), diagnosed on Asia 1997 and Argentina 2001. **Globalization’s structure: value chains** (TiVA, dependency, chokepoints) and their fragility under COVID, war, and export controls; and the **elephant curve**, whose stagnant rich-country trough is the Stolper–Samuelson loser and the engine of the trade backlash — now playing out in imbalances, reshoring, and the weaponisation of the dollar and the chip. Throughout, the discipline is the same: name the mechanism, add up the winners and losers, and state what the model cannot settle.

### Exercises

1. Two countries have the unit-labour-requirement table of Section 1.1 (Home: 1 hr cloth, 2 hr wine; Foreign: 6 hr cloth, 3 hr wine). Identify each country’s comparative advantage from opportunity costs, state the range of world relative prices at which both gain, and compute Home’s gain from trade at a world price of 1 wine per cloth. Explain why Home gains **despite** having an absolute advantage in both goods.
2. State the Heckscher–Ohlin theorem and the Stolper–Samuelson corollary. For a skill-abundant advanced economy opening to trade with a labour-abundant economy, identify the factor that gains and the factor that loses in **real** terms, and explain precisely why “the country gains from trade” and “every citizen gains from trade” are different claims — naming the policy that would reconcile them.
3. Explain why Germany and France trade cars for cars, which classical model fails to predict this, and which mechanism (scale economies, product differentiation) New trade theory uses instead. State the source of the gains, and explain why intra- industry trade is politically easier than inter-industry trade.
4. A small country has demand  $P = 10 - Q$  and supply  $P = Q$ ; the world price is 2 and a tariff of 2 is imposed. Compute the new internal price, domestic supply, demand and imports; the producer-surplus gain, tariff revenue, consumer-surplus loss, and the two deadweight-loss triangles; and the net national welfare change. Then explain how the verdict differs for a **large** country (terms of trade) and why retaliation undoes the gain.
5. Compare a **tariff**, an import **quota**, and an **export subsidy** on welfare grounds, identifying who captures the equivalent of the tariff-revenue rectangle in each case. Then

evaluate the **infant-industry** and **national-security** arguments for protection, stating the precise condition under which each survives a cost–benefit test.

**6.** Choose **one** of: the US–China tariff war (2018–2025), the US Inflation Reduction Act, or the EU CBAM. Using the course’s tools, classify the instrument, name the winners and losers on explicit mechanisms, identify any externality, and deliver a net-welfare verdict — explaining why the same instrument (a tariff) earns opposite verdicts in the US–China and CBAM cases.

**7.** Write the balance-of-payments identity and explain why a current-account deficit is by construction a financial-account surplus. Using  $CA = S - I$ , give one interpretation under which a deficit signals strength and one under which it signals weakness, and explain what additional information settles which applies.

**8.** A Big Mac costs \$5.00 in the US and £4.00 in the UK; the market rate is \$1.40 per pound. Compute the PPP-implied rate and state whether the pound is over- or under-valued. Then, given a euro rate of 3% and a dollar rate of 5%, use UIP to find the market’s expected dollar appreciation or depreciation. Explain how PPP and UIP can predict opposite movements at once (overshooting).

**9.** State the Mundell–Fleming trilemma and draw the triangle, labelling which goal each feasible regime sacrifices. Classify the Eurozone, China’s RMB management, and a freely floating emerging economy by the corner each gives up, and explain why a fixed-rate country with open capital markets cannot run an independent monetary policy.

**10.** Define the first-, second-, and third-generation currency-crisis models, naming the cause in each. Diagnose the 1997 Asian crisis and the 2001 Argentina episode, saying which mechanism dominates each, and explain how the trilemma accounts for why each country was exposed in the first place.

**11.** Explain why gross trade statistics misattribute the value of a smartphone and how OECD TiVA corrects this, defining backward and forward linkages. Use them to explain how dependency mapping locates a supply-chain chokepoint, and assess the resilience-versus-efficiency trade-off of reshoring in light of Chapter 1’s gains from trade.

**12.** Interpret Milanovic’s elephant curve: identify the three constituencies behind its hump, trough, and trunk, and connect the rich-country trough to the Stolper–Samuelson theorem of Chapter 1. Explain why the trade backlash is concentrated in rich-country electorates even though global between-country inequality fell, and why economic weaponisation (dollar primacy, export controls) is self-eroding.